

Data Structures Lab

Laboratory File • Spring Semester 2026



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Submitted To **Ahsan Ul Haq**

Programme **B.Tech – CSE**

Session **Spring 2026**

Problem 1 - Hot Potato Game [Circular Queue]	
Problem 2 – Bank Teller Service Simulation [Queue Application]	
Problem 3 – Reverse First K elements of a Queue [Queue+Stack]	
Problem 4 – Queue using Two Stacks [ADT Design])	

— *Spring Session 2026* —

About This Lab File

Purpose of This Lab File

This lab file documents **all experiments** performed throughout the **Data Structures Lab** course (4th Semester, B.Tech CSE) at NIT Srinagar during the **Spring 2026 session**. Topics covered across the semester include Arrays, Searching, Sorting, Linked Lists, Stacks, Queues, Trees, Graphs, and Hashing.

Each experiment is recorded in a structured format:

- **Problem Statement** – What we are solving
- **Pseudocode** – Step-by-step algorithm logic
- **C++ Solution** – Complete working code
- **Program Output** – Expected Output
- **Complexity Analysis** – Time & Space analysis
- **Key Takeaways** – What we learned
- **Program Screenshot** – Verified test results